EN3160 Assignment 1 on Intensity Transformations and Neighborhood Filtering

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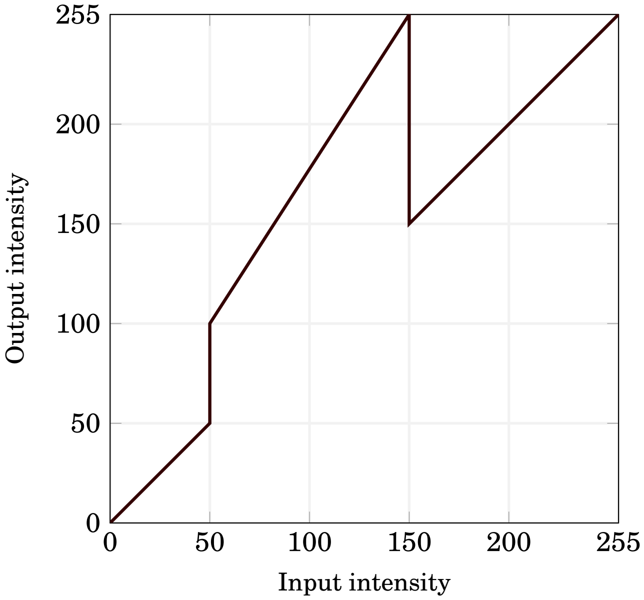
September 2, 2026

1. Implement the intensity transformation depicted in Fig. 1a on the image shown in Fig. 1b by writing a function intensity_transform(im, breakpoints) where the break points are, e.g.,

$$\begin{bmatrix} 0 & 0 \\ 50 & 50 \\ 100 & 150 \\ 150 & 255 \\ 255 & 255 \end{bmatrix}.$$

Experiment with different break points to get a visually pleasing output. Show the intensity transformation as a plot and the original and transformed images side-by-side.



(a) Intensity transformation.

(b) Image for intensity transformation.

2. Apply a similar operation as above (question 1) to accentuate
   (a) white matter
   (b) gray matter

   in the brain proton density image shown in Fig. 2. Show the intensity transformations as a plots.

3. Consider the image shown in Fig. 3[1].
   (a) Apply gamma correction to the L plane in the $L^*a^*b^*$ color space and state the γ value.
   (b) Show the histograms of the original and corrected images.

[1] https://www.adobe.com/creativecloud/photography/discover/highlights-and-shadows.html

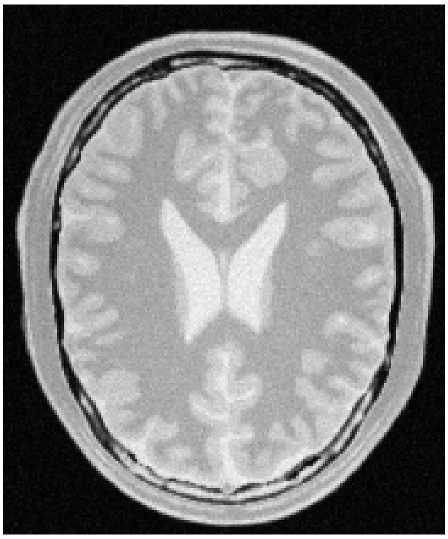


Figure 2: A brain proton density slice.

Figure 3: Image for gamma correction.

4. Increasing the vibrance of a photograph is probably achieved by applying an intensity transformation such as
$$f(x) = \min\left(x + a \times 128e^{-\frac{(x-128)^2}{2\sigma^2}}, 255\right),$$
to the saturation plane, where x is the input intensity, $a \in [0,1]$ and $\sigma = 70$.
   (a) Split the image shown in Fig. 4 into hue, saturation, and value planes.
   (b) Apply the aforementioned intensity transformation to the saturation plane.
   (c) Adjust a to get a visually pleasing output. Report the value of a.
   (d) Recombine the three planes.
   (e) Display the original image, vibrance-enhanced image, and the intensity transformation.

Figure 4: Image for enhancing the vibrance.

5. Write a function of your own to carry out histogram equalization on the image shown in Fig. 5. Show the histograms before and after equalization.

6. In this question, we will apply histogram equalization only to the foreground of an image to produce an image with a histogram equalized foreground.

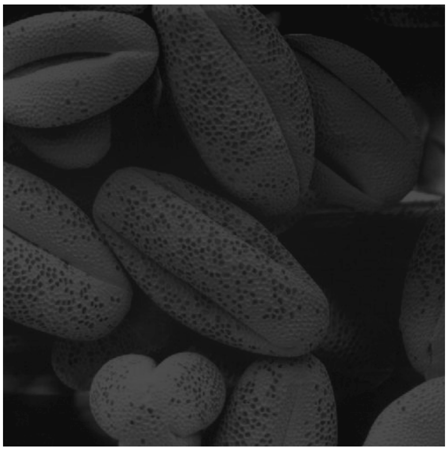


Figure 5: Image for histogram equalization.

(a) Open the image in Fig. 6, split it into hue, saturation, and values and display these planes in grayscale.

(b) Select the appropriate plane to threshold in extract the foreground mask. A mask is a binary image.

(c) Now obtain the foreground only using cv.bitwise_and and compute the histogram.

(d) Obtain the cumulative sum of the histogram using np.cumsum.

(e) Use the formulas in slides to histogram-equalize the foreground.

(f) Extract the background and add with the histogram equalized foreground.

Show the hue, saturation, and value plane, the mask, the original image, and the result with the histogram-equalized foreground.

Figure 6: Image for histogram equalizing the foreground.

7. Filtering with the Sobel operator can compute the gradient. Consider the image shown in Fig. 7

   (a) Using the existing filter2D to Sobel filter the image.

   (b) Write your own code to Sobel filter the image.

   (c) Using the property

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \end{bmatrix},$$

   carry out Sobel filtering.

8. Write a program to zoom images by a given factor $s \in (0, 10]$. You must use a function to zoom the image, which can handle

   (a) nearest-neighbor, and

   (b) bilinear interpolation.

   I have included four images, two large originals, and there zoomed-out versions. Test you algorithm by computing the normalized sum of squared difference (SSD) when you scale-up the given small images by a factor of 4 by comparing with the original images.

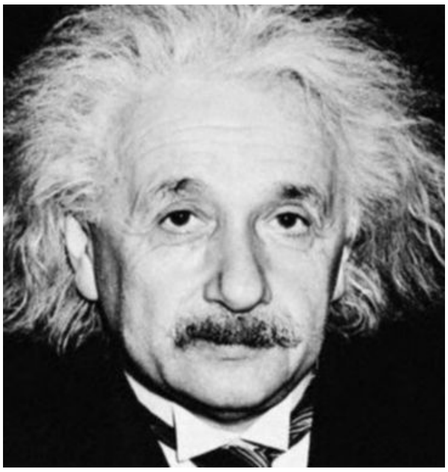


Figure 7: Image for Sobel filtering.

9. Fig. 8[2] shows a flower image with both the foreground and background are in focus.

   (a) Use grabCut to segment the image. Show the final segmentation mask, foreground image, and background image.

   (b) Produce an enhanced image with a substantially blurred background. Display the original image alongside the enhanced image.

   (c) Why is the background just beyond the edge of the flower quite dark in the enhanced image?

Figure 8: Image enhancing.

10. Fig. 9 shows an image of Lake Minaret. The reuirement is to smoothen the image while keeping the edges sharp.

    (a) Use OpenCV's bilateralFilter to smoothen the image with a bilateral filter with σ_s and σ_r of your choice. Display the original, Gaussian blured image with a similar kernel, and bilateral filtered images side-by-side.

    (b) Implement your own bilateral filter and compare the results with OpenCV's bilateralFilter. Display the original, Gaussian blured image with a similar kernel, and bilateral filtered images side-by-side. Use a suitable quantitative measure to compare the results.

[2] https://steemit.com/marguerite/ctrl-alt-nwo/marguerite-daisy

Figure 9: Image bilateral filtering.

GitHub Profile

You must include the link to your GitHub (or some other SVN) profile, so that I can see that you have worked on this assignment over a reasonable duration. Therefore, make commits regularly.

Submission

The report must be a PDF exported directly from your Jupyter Notebook. Reports created by manually copying and pasting code snippets and images into another document will not be accepted. Upload a report (maximum 10 pages) named `your_index_a01.pdf`. Include your index number and name within the PDF as well. The report should contain the key parts of your code, representative image results, comparisons of results, and a clear interpretation and discussion of the findings. The interpretation of the results and the discussion are important components of the report. An extra-page penalty of 20 marks per page will be applied for pages beyond the 10-page limit.

Grading policy: This assignment contributes **5%** to the final grade. Since LLMs can be used to generate code, you are required to include a link to your GitHub profile in the report. We will review your commit history and timestamps to verify that the assignment was completed over a reasonable period. Therefore, commit your work regularly throughout the assignment. For efficiency, only the submitted PDF will be graded. However, your thorough and detailed understanding of the assignment will be assessed in the final examination through questions about your implementation and results.